DATA SCIENCE AND MACHINE LEARNING IN NATURAL SCIENCE - AUTUMN 2026

Mid-Way Project Check

Overall Instructions

> ! **Deadline: 3rd of November, 8:00AM**
>
> Submit a reflection by the **3rd of November** (Masters: one per group; PhD: individual).

Submission Requirements:

- **Length:** Minimum 2, maximum 4 pages (excluding references)
- **Writing tone:** Write as if to a peer in this course
- **File format:** PDF
- **Naming convention:** `midway-check-<lastname-or-groupname>.pdf`
- Submit the document through Canvas

> i **Core Requirement**
>
> The purpose of this checkpoint is to evaluate how your project is progressing since the initial abstract. We are particularly interested in what you have **actually implemented and learned so far**, whether your original problem remains feasible, and whether you have a clear path toward the final project.
> This is not expected to be a polished final report. Preliminary or unsuccessful results are completely acceptable, provided that you can interpret them and explain what they imply for the next stage of the project.

Required Content (Keep it concise)

1. Problem and Current Project Scope

Briefly restate the problem you are trying to solve. Do not repeat the full motivation from your abstract. Instead, focus on the current formulation of the problem.

- What is your current research question or prediction task?
- What are the model inputs and outputs/targets?
- Has the problem changed since your original abstract? If so, explain briefly what changed and why.

2. Data and Progress So Far

Describe the data you are actually working with and what you have done with it.

- What dataset(s) are you currently using?
- How many observations, features, classes, time series, images, etc. are available?
- What data cleaning, preprocessing, feature construction, exploratory analysis, or train/validation/test splitting have you completed?
- Have you discovered any important limitations in the data?

Focus on the data you have obtained rather than data you still hope to obtain.

3. Methods and Baseline

Explain what you have implemented so far.

- What baseline have you established?
- Why is this an appropriate baseline for your problem?
- What machine-learning method(s) have you tested so far?
- What evaluation metric(s) are you using, and why?

The baseline should provide a meaningful reference against which your machine-learning model can be evaluated. Depending on the problem, this may be a simple statistical model, a naive predictor, an established physical or empirical model, or a method reported in the literature.

4. Preliminary Results

Present your most informative results so far.

- Report the performance of your baseline.
- Report the performance of your current machine-learning model(s).
- Include at least one useful quantitative comparison, table, or figure where appropriate.
- Clearly distinguish training performance from validation/test performance.

If your current method does not outperform the baseline, this is not by itself a problem. The important part is that you investigate and discuss why.

5. Interpretation and Feasibility

Based on your work so far, critically evaluate the current state of the project.

- Is your original problem still feasible within the remaining project time?
- Are your models learning something useful, or are there signs of overfitting, data leakage, insufficient data, weak features, class imbalance, or other limitations?
- Do you expect the current approach to solve the problem adequately, or does the problem formulation or methodology need to change?

6. Next Steps

Explain what you will do between this checkpoint and the final project report.

- If you already outperform your baseline, what will you investigate next?
- If you do not yet outperform the baseline, what specific changes do you expect could improve the result?
- Are you planning changes to the features, preprocessing, model family, hyperparameters, evaluation procedure, or problem formulation?
- What is the main remaining technical risk, and what will you do if your preferred approach does not work?

References

Include the most relevant references for your dataset, methodology, baseline, or comparison with previous work. Keep the reference list concise.